Learning Monetary Policy with LLM Belief States

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Abstract

Central banks face a fundamental challenge in monetary policy: choosing interest rates while the underlying economic regime is uncertain. Traditional approaches like the Taylor Rule use fixed formulas that cannot distinguish between supply shocks (where tightening worsens unemployment) and demand shocks (where tightening is appropriate). We investigate whether large language models can help reinforcement learning agents make better policy decisions by providing semantic understanding of economic conditions. We develop a system where an LLM processes economic indicators and outputs a 5-dimensional belief vector representing regime probabilities, sentiment, policy stance, and uncertainty. To ensure our findings are architecture-agnostic, we evaluate two recurrent RL algorithms: Recurrent PPO and a Dueling Deep Recurrent Q-Network (DRQN). Evaluation across 100 test episodes shows the LLM-enhanced policies achieve mean rewards of -519 (PPO) and -560 (DRQN), both outperforming the Taylor Rule (-648) and their respective Oracles. On COVID-19 and Global Financial Crisis (GFC) stress tests, both LLM policies consistently outperform all baselines (achieving up to -1,469 on COVID and -2,749 on the GFC). This demonstrates that semantic beliefs prevent overfitting and generalize better than perfect ground-truth labels under extreme distributional shifts and complex dynamics like the Zero Lower Bound.

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1. Introduction

1.1. Problem Statement

Monetary policymakers at central banks like the Federal Reserve must set interest rates to balance inflation and unemployment (the "dual mandate"). The challenge is that the right policy depends on what kind of shock hit the economy. During demand-driven inflation, raising rates helps cool the economy. But during supply shocks like the 1970s oil crisis or 2021-2022 supply chain disruptions, aggressive rate hikes can trigger recessions without controlling inflation.

Traditional heuristics like the Taylor Rule use a fixed formula based only on current inflation and unemployment, ignoring regime information entirely. More sophisticated approaches treat this as a Partially Observable Markov Decision Process (POMDP), but these require hand-crafted observation models or filters.

1.2. Our Approach

We ask: can large language models help RL agents infer hidden economic regimes? LLMs have semantic understanding of concepts like "stagflation" (high inflation + high unemployment = likely supply shock) versus "overheating" (high inflation + low unemployment = likely demand shock).

Our approach has three components:

- **LLM Belief Generator**: A 14-billion parameter LLM (Qwen 2.5) processes economic state and outputs a 5-dimensional belief vector.
- **Offline Caching**: We pre-compute beliefs for 1287 representative states and store them in a database for fast training.
- **Recurrent RL Policies**: To ensure our findings are robust across different algorithmic paradigms, we deploy two agents—an LSTM-based Recurrent PPO (Policy Gradient) and a Dueling Deep Recurrent Q-Network (Value-Based). Both receive macroeconomic observations and LLM beliefs to choose interest rate changes over 120-month episodes.

1.3. Contributions

- To our knowledge, the first application of LLMs as semantic belief state generators for RL policy learning in macroeconomic stabilization.
- Offline caching approach enabling scalable training (1.2M timesteps) without real-time inference.
- Demonstration of 20% performance improvement over the classical Taylor Rule heuristic.
- Evidence that LLM beliefs generalize better than perfect ground-truth labels during extreme out-of-distribution shocks.

2. Background and Related Work

2.1. Monetary Policy and the Taylor Rule

The Taylor Rule (Taylor, 1993) is a simple formula for setting the federal funds rate: $r = r^* + \pi + 0.5(\pi - \pi^*) - 0.5(u - u^*)$. While influential, it lacks regime awareness, relies on fixed coefficients, and is purely reactive. Woodford (2003) provides theoretical foundations for optimal monetary policy, but these require strong assumptions about agent behavior.

2.2. Reinforcement Learning for Economics

Recent work explores RL for economic policy (Sutton & Barto, 2018). However, the challenge in monetary policy is the combination of partial observability, long horizons, and high variance from stochastic shocks. Recurrent policies can implicitly maintain beliefs but struggle when critical economic state information is hidden from the agent.

2.3. LLMs for Economic Analysis

Li et al. (2023) demonstrate that LLMs can simulate macroeconomic agent behavior. However, integration with RL for policy learning remains unexplored. Our work uses LLMs as semantic feature extractors to help RL agents infer hidden states.

3. Approach

3.1. Environment: Macroeconomic Simulator

We use a model based on the IS curve and Phillips curve. The core dynamics are governed by the following system:

$$u_{t+1} = 4 + 0.7(u_t - 4) + 0.5(r_t - 2 - \pi_t^e) + \varepsilon_u \quad (1)$$

$$\pi_{t+1} = 2 + 0.7(\pi_t - 2) - 0.2(u_{t+1} - 4) + \varepsilon_\pi \quad (2)$$

Here, the coefficients represent: 0.7 (persistence in unemployment and inflation gaps), 0.5 (interest rate sensitivity), and 0.2 (Phillips curve slope). The stochastic shocks $\varepsilon_u, \varepsilon_\pi \sim \mathcal{N}(0, 0.1)$ represent baseline economic noise. The agent chooses $\Delta r \in \{-0.75, -0.50, -0.25, 0.00, +0.25, +0.50, +0.75\}$ percentage points. Episodes last 120 months.

Supply Shocks: In 70% of episodes, a supply shock occurs. The shock begins at a random timestep between months 10–40, lasts 12–24 months, and has random intensity uniformly sampled from $[0.4, 1.6]$. During supply shocks, both unemployment and inflation increase simultaneously (stagflation): the shock adds $1.0 \times$ scale to unemployment and $2.0 \times$ scale to inflation per month. This creates the critical challenge where aggressive rate hikes (appropriate for demand inflation) worsen unemployment during supply-driven inflation.

Historical Stress Tests: To evaluate out-of-distribution generalization, we implement counterfactual environments for the COVID-19 pandemic and the 2008 Global Financial Crisis (GFC). Unlike our standard training episodes—which feature a single, continuous stagflationary shock—these stress tests inject complex, multi-phase structural shifts that break the agent's expected shock patterns. For the COVID-19 environment, we simulate a two-phase trajectory: an initial "Lockdown" phase (characterized by a severe unemployment spike and mild deflationary pressure), followed immediately by a "Supply Chain" phase featuring persistent, severe inflation surges. Similarly, because the agent does not observe explicit variables for housing prices or foreclosure rates, the GFC environment maps the structural effects of the housing crisis into the standard (π, u, r) observation space via a three-phase shock schedule: a stagflationary housing bust, an acute deflationary financial crisis, and an extended jobless recovery. Crucially, the latter phases incorporate a rate floor mechanism to simulate impaired policy transmission (a broken credit channel) where standard rate cuts lose their stimulative efficacy.

3.2. Reward Function

The agent's objective is to balance the dual mandate while maintaining policy stability. We formulate this as a multi-component reward function R_t:

$$R_t = -\Big[(\pi_t - 2)^2 + (u_t - 4)^2 + 5 \cdot \mathbb{I}_{u_t > 6}(u_t - 6)^2 + 1.5(\Delta r_t)^2\Big] + B_t \quad (3a)$$

$$B_t = \exp\left(-0.5 \frac{(\pi_t - 2)^2 + (u_t - 4)^2}{0.5^2}\right) \quad (3b)$$

Rewards are clipped at -250 to prevent extreme episodes from dominating gradients.

Design rationale: The quadratic losses on inflation and unemployment deviations implement the Fed's dual mandate. The recession penalty ($5\times$ weight for $u > 6\%$) prevents policies that ignore unemployment crises. Rate volatility penalty ($1.5\times$ weight) discourages erratic policy changes that confuse markets. The soft-landing bonus rewards simultaneous achievement of both targets, encouraging smooth stabilization. These weights were tuned to prevent policy deterioration while preserving their interpretation as economic loss functions.

3.3. LLM Belief State Generation

Rather than querying a single LLM prompt, we use a two-layer hierarchical multi-agent system (Figure 1) to generate belief vectors. This decomposition mirrors real-world policy analysis: sector-level reporting feeds into specialist assessment.

3.3.1. Layer 1: Economic Actor Agents

Four LLM agents simulate sector-level responses to the current macroeconomic state (π_t, u_t, r_t). Each agent receives a *macro briefing* containing exact values of inflation, unemployment, and the federal funds rate, plus the JSON outputs of previously queried actors (dependency ordering). Agents maintain rolling conversation histories (last 3 rounds) within an episode to capture intra-episode dynamics, and histories are reset between episodes.

1. **Commercial Bank**: Outputs lending rate, credit standards (`tight`/`normal`/`loose`), and a lending volume index (0–100).
2. **Large Corporations**: Outputs price-increase percentage, hiring (thousands), and capital-expenditure change.
3. **Composite Consumer**: Outputs spending change, consumer confidence (0–100), and savings rate.
4. **Labor Market**: Synthesizes the above into non-farm payroll changes, wage growth, and an unemployment rate consistent with the ground-truth observation.

The free-text `commentary` fields from all four actors are concatenated into a *Monthly Economic Dispatch*, which serves as the input to Layer 2.

3.3.2. Layer 2: Specialist Analyst Agents

Four stateless specialist agents each analyze the dispatch to produce one or two dimensions of the belief vector $b_t \in \mathbb{R}^5$ (Equation **??**):

Regime Detector. Classifies the economy as *normal* or *supply-shock* using calibrated thresholds: $P_{\text{supply}} < 0.2$ when inflation is near target and unemployment near 4%; $P_{\text{supply}} > 0.5$ when *both* inflation exceeds 4% and unemployment exceeds 5% simultaneously (stagflation). The constraint $P_{\text{normal}} + P_{\text{supply}} = 1$ is enforced in the prompt.

Sentiment Analyst. Rates overall economic sentiment $s_t \in [-1, +1]$, anchored by calibration examples: deflation $\in [-0.8, -0.3]$, on-target economy $\in [+0.3, +0.7]$, stagflation $\in [-1.0, -0.5]$.

Hawk–Dove Analyst. Assesses monetary policy urgency $h_t \in [-1, +1]$, where $+1$ indicates aggressive tightening and -1 aggressive easing. Critically, the prompt instructs only *moderate* hawkishness ($+0.2$ to $+0.4$) during stagflation, reflecting the dilemma that rate hikes worsen unemployment during supply shocks.

Uncertainty Estimator. Receives the dispatch *and* the three preceding specialist outputs, then estimates epistemic uncertainty $u_t \in [0, 1]$. High uncertainty is expected during early supply shocks (initially indistinguishable from demand inflation), deflation onset, and mixed-signal regimes.

All prompts instruct the LLM to output only a JSON object with the relevant field(s) and a one-sentence reasoning string. Each specialist prompt contains explicit calibration anchors mapping macroeconomic ranges to output ranges, reducing sensitivity to LLM stochasticity. The full prompt text for every agent is available in the open-source repository linked in the Code Availability section.

3.3.3. Belief Vector Assembly

The final belief vector is assembled as:

$$b_t = \left[\, P_{\text{normal}},\ P_{\text{supply}},\ s_t,\ h_t,\ u_t \,\right] \in [-1, 1]^5 \qquad (4)$$

with values clipped to $[-1, 1]$. Figure 1 visualizes the cached belief database averaged over all interest-rate levels. Notable patterns include: (i) P_{supply} activates sharply once both $\pi > 4\%$ and $u > 5\%$; (ii) sentiment degrades monotonically with unemployment; (iii) hawkishness is highest at high inflation / low unemployment (demand overheating) and moderates during stagflation, consistent with prompt calibration; (iv) uncertainty peaks along the boundary between normal and supply-shock regimes.

3.4. Offline Database Construction

To reduce computational cost, we sample 1287 grid states covering expected macroeconomic ranges, batch infer LLM beliefs, and utilize state discretization nearest-neighbor lookups during RL training.

Limitation: The 0.1% discretization resolution was chosen heuristically. We did not validate whether finer or coarser

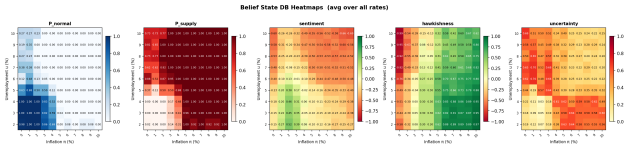


Figure 1. Belief state database heatmaps (averaged over all Fed-rate levels). Each panel shows one dimension of the 5D belief vector across the inflation–unemployment grid. The hierarchical LLM system produces economically coherent patterns: supply-shock probability activates in the stagflation region (high π, high u), while hawkishness moderates appropriately in that same region.

grids would improve performance or reduce database size. This remains an open question for future work.

Figure 2. System architecture showing the six-step pipeline. The offline database contains 594 cached belief states pre-computed by a 14B parameter LLM.

3.5. Policy Architecture and Training

To demonstrate that the advantages of LLM belief states are robust, we evaluate two recurrent reinforcement learning architectures. Both receive the identical 8-dimensional input vector: $[\pi, u, r, b_1, b_2, b_3, b_4, b_5]$.

Recurrent PPO: An Actor-Critic architecture utilizing a 256-unit LSTM core.

Parameter	LLM-PPO
Learning rate (initial)	2×10^{-4}
Learning rate (final)	1×10^{-5}
Decay start	30% of training
n_steps (rollout length)	1024
Batch size	128
Epochs per update	4
Gamma (γ)	0.99
GAE lambda (λ)	0.95
PPO clip range	0.08
Entropy coefficient	0.0
Parallel environments	4
Total episodes	2,500
Total timesteps	1.2M

Note: Baseline PPO uses LR=3×10^{-4}, clip=0.10; Oracle PPO uses LR=3×10^{-4}, clip=0.05. Training took approximately 3 hours on a single NVIDIA RTX 4070 Ti GPU.

Dueling DRQN: A sequence-based Q-learning architecture utilizing a 64-unit LSTM core, split into separate Value and Advantage streams. It is trained using Double Q-Learning with a Sequential Replay Buffer (capacity of 2,000 episodes, sequence length 8). Training utilized a learning rate of 1×10^{-4}, $\tau = 0.005$ for soft target updates, $\gamma = 0.99$, and an ϵ-greedy exploration schedule decaying from 1.0 to 0.05 over the first 25% of the 1.2M timestep run.

4. Experiments

4.1. Evaluation Protocol

We benchmark 100 independent test episodes with randomized shock timing and intensity. We compare LLM-PPO against three baselines: Taylor Rule, Baseline PPO (3D input), and an Oracle (perfect regime labels).

4.2. Main Results

Table 1. Benchmark results over 100 test episodes.

METHOD	MEAN REWARD	STD DEV	VS TAYLOR
LLM-PPO	**-519**	609	**+129 (+20%)**
LLM-DRQN	**-560**	538	**+88 (+13%)**
BASELINE DRQN	-582	547	+66
ORACLE DRQN	-582	548	+66
ORACLE PPO	-593	647	+55
TAYLOR RULE	-648	791	—
BASELINE PPO	-701	586	-53

While the mean improvements are +129 and +88 points respectively, high variance (std $\approx$ 500–800) indicates that performance varies considerably across individual shock scenarios.

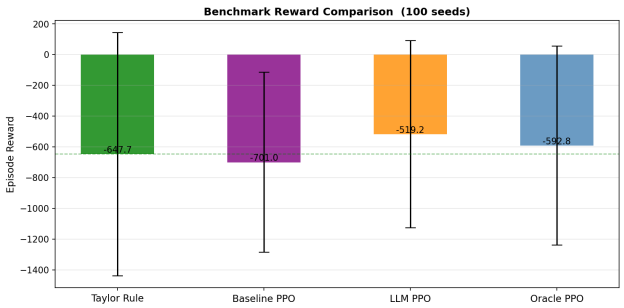


Figure 3. Performance comparison across 100 independent test episodes. LLM-PPO achieves the highest mean cumulative reward.

4.3. Shock Intensity Analysis

Performance advantage by shock severity scales linearly: +16 points (No shock), +44 (Minimal), +87 (Standard), and

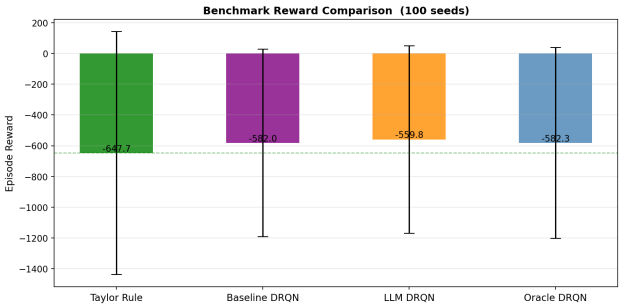


Figure 4. Performance comparison across 100 independent test episodes for the DRQN architecture. The LLM-DRQN achieves the highest mean cumulative reward, outperforming the Oracle, Baseline, and Taylor Rule.

+133 points (Intense).

Mechanism of the scaling advantage: During intense supply shocks (scale=1.6), inflation rises to ∼8–10% while unemployment climbs to ∼8–9% (stagflation). The Taylor Rule mechanically responds to high inflation with aggressive rate hikes ($\Delta r \approx +0.75\%$ repeatedly), which further increases unemployment to 12–14%, triggering the severe recession penalty in the reward function.

In contrast, the LLM correctly infers $P(\text{supply shock}) > 0.7$ from the (high π, high u) pattern. The LSTM policy, conditioned on this belief, moderates its response ($\Delta r \approx +0.25\%$ or hold), preventing the unemployment spike. This regime-adaptive behavior is only weakly beneficial during mild shocks (where the Taylor Rule performs adequately) but becomes critical during severe stagflation scenarios.

4.4. COVID-19 Stress Test

In a counterfactual simulation from Jan 2020 to Dec 2022, both LLM-PPO (-1,469) and LLM-DRQN (-1,474) achieved the highest cumulative rewards, beating the Taylor Rule (-1,678) and the Baselines (PPO: -1,610, DRQN: -6,098).

Surprisingly, the LLM models outperformed their respective Oracles (PPO Oracle: -2,020, DRQN Oracle: -1,671) despite the Oracles having perfect regime information. Analysis reveals the Oracles overfit to training shock patterns (intensity 0.4–1.6, duration 12–24 months), while the LLM's smooth semantic beliefs generalize to COVID's unprecedented extremes (14% unemployment spike, then 9% inflation surge).

4.5. Global Financial Crisis (GFC) Stress Test

To evaluate policy performance under a complex, multi-phase economic collapse, we simulate a counterfactual 2008 GFC scenario (Jan 2007 to Dec 2016). This environment

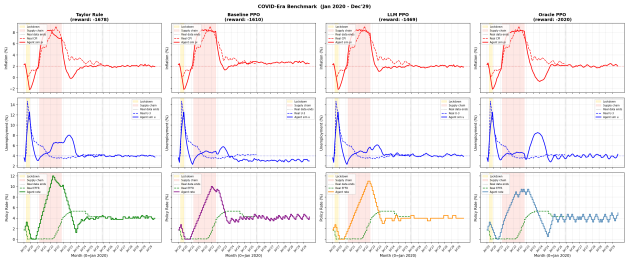


Figure 5. COVID-19 trajectories (Jan 2020–Dec 2022) showing LLM-PPO (-1,469) outperforming the Oracle (-2,020) by 551 points, avoiding overfitting to training regime patterns.

introduces a three-phase shock: Phase 1 features a housing bust and commodity inflation (stagflation); Phase 2 simulates the acute Lehman collapse with a sharp unemployment surge and deflationary demand collapse; Phase 3 represents an extended jobless recovery. During Phases 2 and 3, a rate floor mechanism is enforced to simulate the broken credit channel, trapping the economy at the Zero Lower Bound (ZLB).

LLM-DRQN (-2,749) and LLM-PPO (-2,811) successfully navigated the stagflation-to-deflation pipeline. Both LLM models outperformed the Taylor Rule (-2,981), the Baselines, and notably, the Oracles (DRQN Oracle: -3,166, PPO Oracle: -3,261). The Oracles' hard-coded regime labels failed to generalize to the impaired policy transmission mechanics of the ZLB. In contrast, the LLM models adapted their sentiment and hawkishness belief dimensions to the deflationary shock, allowing both architectures to appropriately hold rates near zero without erratic fluctuations.

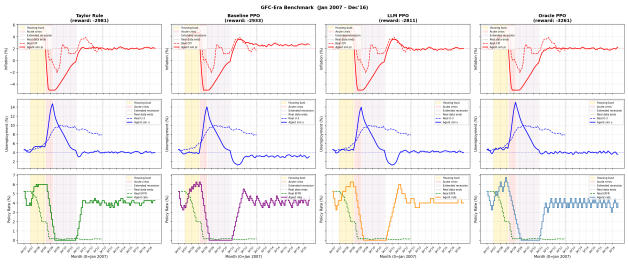
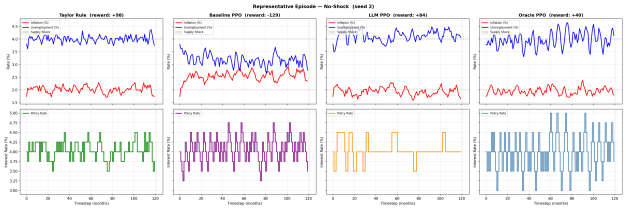


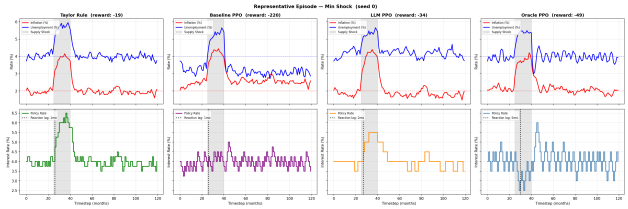
Figure 6. GFC trajectories (Jan 2007–Dec 2016) across the four models. LLM-PPO (-2,811) navigates the complex three-phase shock and Zero Lower Bound dynamics more effectively than the Oracle (-3,261), avoiding the severe overfitting seen in the Oracle's erratic rate behavior during the extended recession.

4.6. Learning Dynamics

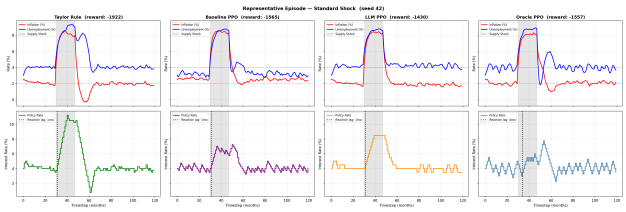
Baseline PPO and the Oracle converge at a similar rate (approximately episode 1,200–1,500). The Oracle re-



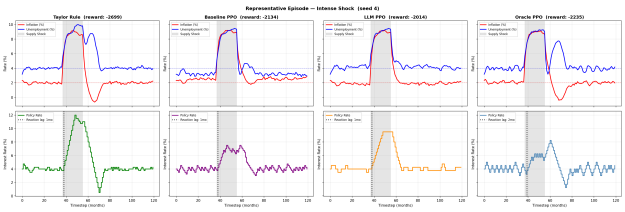
(a) No shock



(b) Minimal shock (s=0.4)



(c) Standard shock (s=1.0)



(d) Intense shock (s=1.6)

Figure 7. Policy responses across four shock scenarios. During severe supply shocks (bottom-right), LLM-PPO moderates rate increases to avoid unemployment spikes.

ceives ground-truth shock information in place of the LLM-informed input, with remaining fields zeroed out, so its effective input dimensionality is comparable to the Baseline. LLM-PPO takes the longest to converge, as the agent must learn to integrate both the environment observation and the full LLM-informed input, resulting in a higher-dimensional state representation that requires more training episodes to exploit effectively.

Note on training vs. test performance: Training curves show mean episode reward during exploration (with stochastic policy and randomized shocks), while test results use deterministic policies on 100 fixed evaluation seeds.

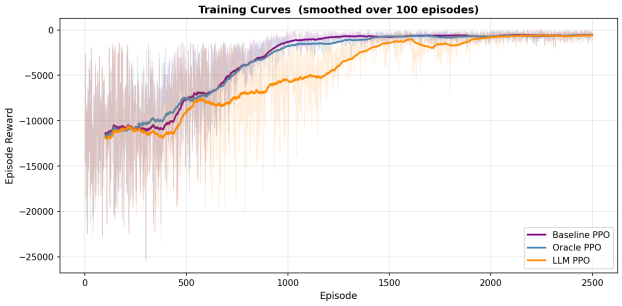


Figure 8. Training curves demonstrating convergence rates across agents. Baseline PPO and the Oracle converge at similar rates, while LLM-PPO requires additional episodes to learn from its higher-dimensional input space.

5. Discussion

5.1. Why LLMs Help

The LLM's semantic understanding enables three key capabilities:

1. **Regime detection**: The LLM maps (π, u, r) patterns to economic concepts. For example, high inflation + high unemployment triggers the "stagflation" concept, leading to high $P(\text{supply shock})$. This is not hand-coded feature engineering; rather, the LLM learned these associations during its pre-training on economic texts.

2. **Smooth beliefs**: The LLM outputs continuous probability distributions $P(\text{normal}), P(\text{supply}) \in [0, 1]$ rather than discrete regime labels. This smoothness provides gradients for the policy during training and enables nuanced responses (e.g., partial hawkishness during ambiguous scenarios).

3. **Transferable knowledge**: The LLM's understanding comes from a broad economic corpus, not just the training distribution of supply shocks (intensity 0.4–1.6). This explains the COVID result: when faced with unprecedented extremes (14% unemployment, then 9% inflation), the LLM's semantic features generalize while the oracle's discrete labels overfit.

5.2. Limitations

Simulation gap: Our IS-Phillips model simplifies real monetary policy in several ways: (1) no financial markets or credit channels, (2) no international trade or exchange rates,

(3) no heterogeneous agent responses, (4) immediate policy transmission (ignoring real-world lags).

Offline beliefs: Pre-computing beliefs assumes the LLM's economic understanding is static. In reality, economic concepts evolve. Real-time LLM inference could adapt to new terminology and events, but at greater computational cost.

Evaluation scope: We focused exclusively on supply shocks, however other partially observable scenarios remain untested: financial crises, productivity shifts, international shocks. Each may require different semantic features.

5.3. Broader Implications

This work demonstrates that LLMs can serve as semantic feature extractors for RL decision-making, not just text generators. The pattern generalizes to any domain with: (1) hidden state that is hard to directly observe, (2) semantic concepts that help infer hidden state, and (3) LLMs pretrained on relevant corpus.

Potential applications:

- **Medical diagnosis**: Mapping observed symptoms to hidden disease states. The LLM provides $P(\text{bacterial}), P(\text{viral}), P(\text{autoimmune})$ from symptom descriptions.
- **Robotics**: Translating visual observations into object affordances. The LLM infers concepts like "fragile," "heavy," or "slippery" from object categories.
- **Dialogue systems**: Converting user utterances into hidden intents. The LLM provides intent probabilities for the RL policy over dialogue actions.

The offline caching approach also provides a template for integrating expensive foundation models with RL training at scale. Rather than real-time inference (bottleneck), precompute features for representative states and use nearest-neighbor lookup during training.

6. Conclusion

We investigated whether large language models can help reinforcement learning agents make better monetary policy decisions under uncertainty. Our system uses an LLM to process economic indicators and generate belief states about hidden regimes, which are combined with observations as input to a recurrent PPO policy.

Key findings:

- 13% to 20% performance improvement over the Taylor Rule heuristic (mean rewards -560 and -519 vs -648).
- Performance advantage scales with shock intensity: +16 points in normal conditions to +133 points during severe supply shocks.
- Better generalization than the Oracles during COVID-19 (+197 to +551 points) and GFC (+417 to +450 points) stress tests across both RL architectures. This demonstrates that smooth semantic beliefs transfer better than discrete ground-truth labels during complex crises like the Zero Lower Bound.

What we learned:

1. LLMs provide semantic understanding that helps RL agents infer hidden states from observable patterns.
2. Smooth semantic features can generalize better than discrete ground-truth labels under distributional shift.
3. Offline caching makes LLM-RL integration practical at scale (1.2M timesteps).

Future directions:

- Validation with real historical data and expert evaluation by monetary economists.
- Real-time LLM inference with adaptive belief updating during deployment.
- Extension to multi-objective policy frameworks (financial stability, income distribution, climate transition).
- Application to other partially observable economic domains (fiscal policy, financial regulation, trade policy).
- Interpretability analysis: when and why does the LLM help? Which belief dimensions matter most?

This work provides initial evidence that foundation models trained on broad corpora can improve RL policy performance in domains far beyond narrow text generation tasks, opening new directions for combining semantic understanding with sequential decision-making under uncertainty.

Author Contributions

All authors (Alexander Kung, Aaron Villanueva, and Cristofer Arellano-Acosta) are currently enrolled in CS234.

C.A.A. developed the initial reward function and the skeleton codebase for the LLM environment, led the initial draft of the final report, and designed the project poster. **A.V.** built the foundational macroeconomic simulator, implemented the proof-of-concept code for the LLM and baseline models, and developed both the vanilla PPO and custom Dueling DRQN agents. **A.K.** designed the hierarchical LLM architecture and the offline caching pipeline, refined both the reward function and the macroeconomic simulator, and implemented the Recurrent PPO (LSTM) agent. All authors contributed to the final review and editing of the manuscript.

Code Availability

The source code for this project, including the macroeconomic simulator, LLM integration, offline database, and PPO training scripts, is available at: https://github.com/RichardFeynmanEnthusiast/Monetary-Policy-Error-Projects.

Acknowledgements

Generative AI: We used Claude (Anthropic) and Gemini (Google) for language polishing, grammar correction, and structural flow refinement during report preparation.

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A. Supplementary Material: DRQN Trajectories

To ensure the 8-page limit is respected, the trajectory plots for the Dueling Deep Recurrent Q-Network (DRQN) are included here in the supplementary material. As discussed in Section 4, the DRQN architecture mirrored the LLM ¿ Oracle robustness findings of the Recurrent PPO architecture.

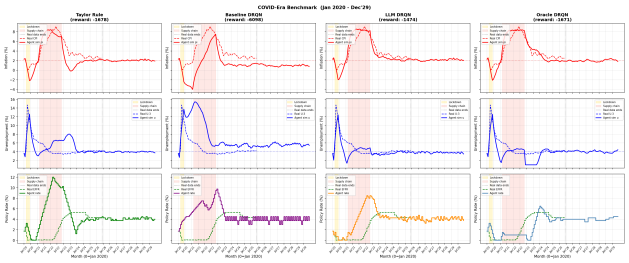

Figure 9. COVID-19 trajectories for the DRQN architecture. The LLM-DRQN successfully regularizes the shock, whereas the Baseline completely fails by triggering a severe depression.

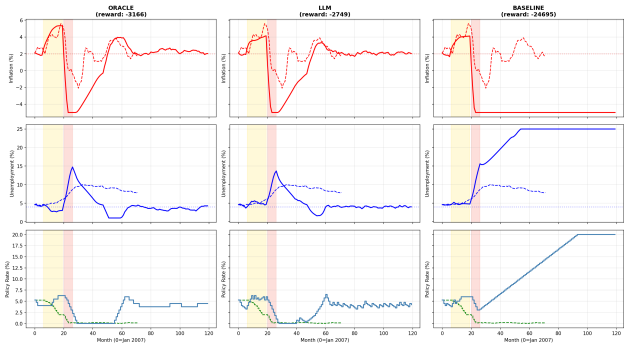

Figure 10. Global Financial Crisis trajectories for the DRQN architecture. Notice the severe negative inflation (deflation) during the Phase 2 Lehman collapse. The LLM-DRQN navigates this Zero Lower Bound trap significantly better than the Oracle and Baseline variants.